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Bioefficiency

Database Set-up, Data Collection and Evaluation (D5.1)

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Abbreviations and Acronyms

BSG	Brewery Spent Grains
CG	Coffee Grounds
DIG	Digestate
EFB	Empty Fruit Bunch
FBW	Fir Brushwood
HM	Horse Manure
HTC	Hydrothermal Carbonisation
ICP-OES	Inductively Coupled Plasma Optical Emission Spectrometry
ICP-MS	Inductively Coupled Plasma Mass Spectrometry
LC	Lawn Cuttings
MIS	Miscanthus
OP	Olive Pomace
SB	Spruce Bark
SE	Steam Explosion
T	Torrefied
ULL	Urban Leaf Litter
STR	Wheat Straw
W	Washed
WP	White Pellets
WPA	White Pellets + Fly Ash
WT	Washed and torrefied
XRD	X-Ray Diffraction
XRF	X-Ray Fluorescence

1. Task Description

The Bioefficiency project aims to advance the general knowledge on biomass combustion and ash-related challenges. To ensure an effective communication and sharing of data between the project partners, an internal database was set up. In this database, all partners can contribute their data and in turn access all data that were provided by the other partners in a simple way.

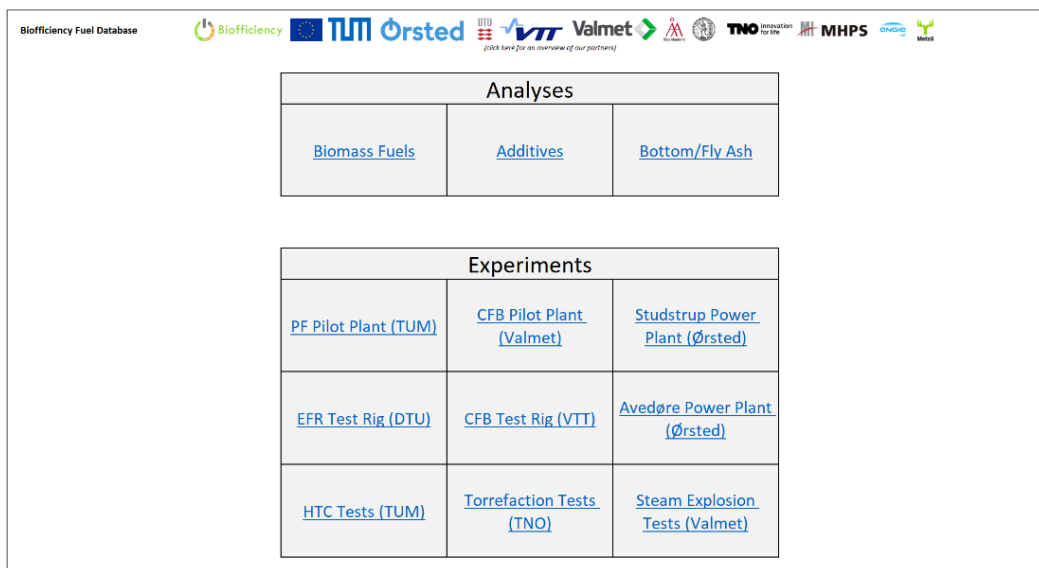
Having all analysis data at one central location also allows for an easier analysis of all data, making it possible to find correlations and outliers, and possibly even catching measurement errors. Finally, having all data centralised makes it easy to publish this data to the public, and therefore serving the goal of promoting and supporting the use of renewable biomass worldwide.

2. Database Design and Setup

In order to ensure that every partner can access all data, it was decided to use Microsoft Excel as the software behind the database. The general setup of the database is as follows: The data is separated into two large categories:

- "Analyses" and
- "Experiments".

In the "Analyses" category, all fuel, additive and ash analyses are integrated, while the "Experiments" category gives an overview of all test rigs, pilot plants and full-scale plants that are used in the Bioefficiency project. In the latter, additional and more detailed experimental data can be shared internally. Figure 1 shows the main page of the database, which includes links to all aforementioned sub-categories as well as links to each partner's homepage.



Analyses		
Biomass Fuels	Additives	Bottom/Fly Ash

Experiments		
PF Pilot Plant (TUM)	CFB Pilot Plant (Valmet)	Studstrup Power Plant (Ørsted)
EFR Test Rig (DTU)	CFB Test Rig (VTT)	Avedøre Power Plant (Ørsted)
HTC Tests (TUM)	Torrefaction Tests (TNO)	Steam Explosion Tests (Valmet)

Figure 1: Overview of the "Start Page" of the Bioefficiency database.

Through the links in Figure 1, the user is directed towards the respective sub-category. Figure 2 shows an example of one of the "Experiments" tabs without the experimental data. The larger part of this database is comprised of fuel and ash analysis data. Selecting one of the respective tabs leads the user to the overview view of all collected data (excerpt see Figure 3). In this overview, it is easy to see which analyses have been carried out on the respective fuel by looking at the name of the partner in each corresponding row. If the user wants to look at any of the present data, the small plus-symbol ("+") on the left side of each window, next to each

analysis, expands the corresponding analysis block and shows the respective analysis data. This expanded view is shown in Figure 4. By implementing the data this way, it is easy to compare two fuels to each other without being overwhelmed by the large amount of total data in the database, while also offering an overview of all available data at a quick glance. Additionally, naming the company that carried out each experiment gives the user a convenient way of knowing which partner to contact if questions arise.

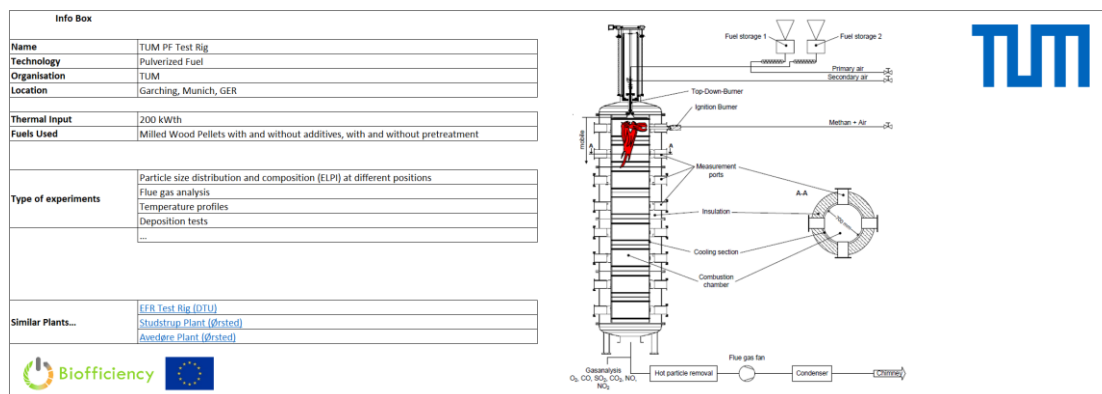


Figure 2: Example of one of the "Experiments" sub-categories.

Biomass Fuel Database		Bioefficiency		
	Fuel Type	Spruce Bark	Spruce Bark Pellets	Spruce Bark
	Pretreated (yes/no, type)	no	no	no
	Additives (yes/no/composition)	no	no	no
	Fuel Provider (Company, Country)	Metsä Group, Finland	Metsä Group, Finland	Metsä Group, Finland
	Date of Analysis	01.02.2017	27.04.2017	04.06.2018
	Name	LY_170201_01	2017/45-2 Val 2 VT 10 (29.06.2016)	Spruce_bark_Valmet
	Basic Analysis	TUM	MHPS	Valmet
	Proximate Analysis	TUM	MHPS	Valmet
	Ultimate Analysis	TUM	MHPS	Valmet
	XRF of Ash	TUM	MHPS	Valmet
	ICP-OES of Ash			Valmet
	ICP-MS of Ash			Valmet
	XRD of Ash			
	Ash Melting Behaviour	TUM	MHPS	
	Particle Size Distribution (Before Grinding)		MHPS	
	Fuel Grinding Behaviour			
	Key Indices, Ratios etc.	automatically	automatically	
	Comments			Actual analysis date was 30.7.-9.8.2018. The date in the "Date of Analysis" row was the sampling date. As received values calculated from originally d.b. results. Analysis subcontracted from Eurofins Labtium Oy, Finland. ICP originally on db fuel.

Figure 3: Overview of three example-fuels in the fuel database.

- Wear rate

All partners submitted analysis and experimental data to TUM, where it was gathered and entered into the database. In the course of the project, the database was continuously updated and uploaded to a shared platform where all partners can access the database anytime.

3.2. Comparison of Different Ash Analyses

As can be seen from the list above, the ash analyses were carried out in multiple different ways. XRF, ICP-OES and ICP-MS are all measuring the general bulk chemical composition of the fuel ashes, leading to possible comparisons between those measurements. In general, both ICP methods are considered superior in resolution and accuracy, not least because XRF analysis assumes each element to be present exclusively in a single oxide state, an assumption that can be highly inaccurate depending on the sample. Nevertheless, if XRF measurements were as accurate as ICP, they could be used in fuel indices that usually rely on accurate elemental composition data from ICP, by calculating the elemental composition of the ash from the oxide state. To investigate the scope in which XRF measurements can still be compared, multiple biomass fuels have been analysed by both XRF and ICP-OES at TUM's internal laboratory. The XRF data (measured in the respective oxide state) was calculated back to the elemental composition and compared to the respective elemental ICP-OES data. An overview over some of the results can be found in Figure 5.

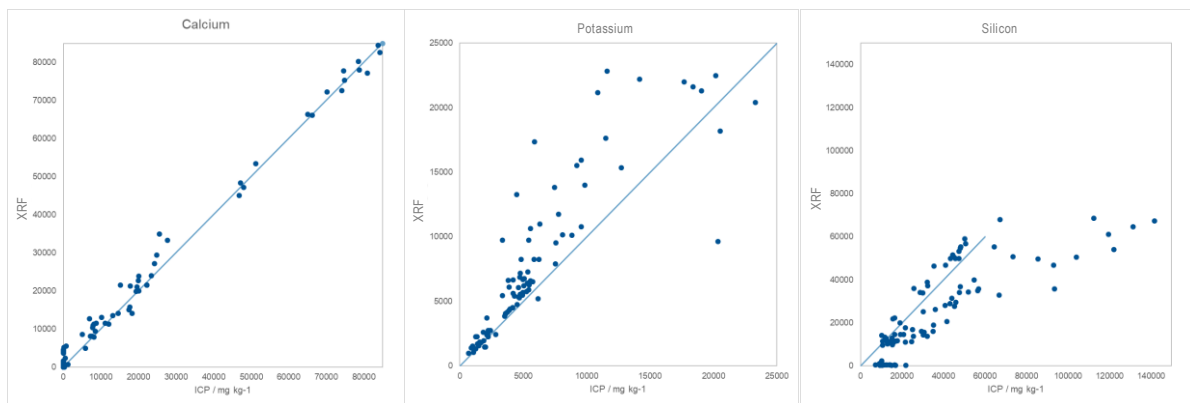


Figure 5: Comparative plot of XRF and ICP-OES data from TUM's laboratory.

Figure 5 shows that while some elements like calcium show comparably good results, other elements like potassium seem to be consistently overpredicted by the XRF measurements. On the other hand, the silicon measurements show a rather even spread up until a certain value of ca. 50.000 mg/kg, after which the linear trend can no longer be observed, hinting at systematic measurement errors for this element in one of the two devices. It should be especially noted though that in the regions of lower concentration, both silicon and calcium show very strong relative deviations towards each other. Since no consistent comparison seems possible for every element, it was decided to not include recalculated XRF data to ICP-OES and ICP-MS data for the calculation of molar indices in the scope of this deliverable.

In general, ICP-MS and ICP-OES measurements show good comparability (see e.g. [1,2]), so that the two methods are assumed to be equivalent in quality. Since most of the database's fuels were investigated either by ICP-MS or ICP-OES, molar indices were calculated with no differentiation between those methods.

3.3. Analysis and Discussion

Overall, 14 different solid biomass fuels have been analysed and entered into the database, ranging from typical wood pellets to herbaceous biomasses, agricultural residues, forestry

residues and low grade residual derived biomasses. When accounting for all pre-treatments and analyses by multiple partners, a total of 57 biomass fuel datasets have been entered.

For a general classification, Figure 6 and Figure 7 show van-Krevelen-diagrams of all the investigated fuels, classifying them by their type and their status of pre-treatment, respectively. When looking at Figure 7, it can be observed that most pre-treatments do not drastically change the H/C or O/C ratio in the fuels, with the exception of HTC treatment: After hydrothermal carbonisation, most fuels see a drastic reduction in hydrogen and oxygen content, leaving a coal-like residue. This also shows in the usual classification categories of the van-Krevelen-diagram, in which the HTC fuels fall into the coal/lignite category. A more detailed analysis of the influence of pre-treatment on the biomasses can be found in the respective Deliverables of the Bioefficiency project [3,4].

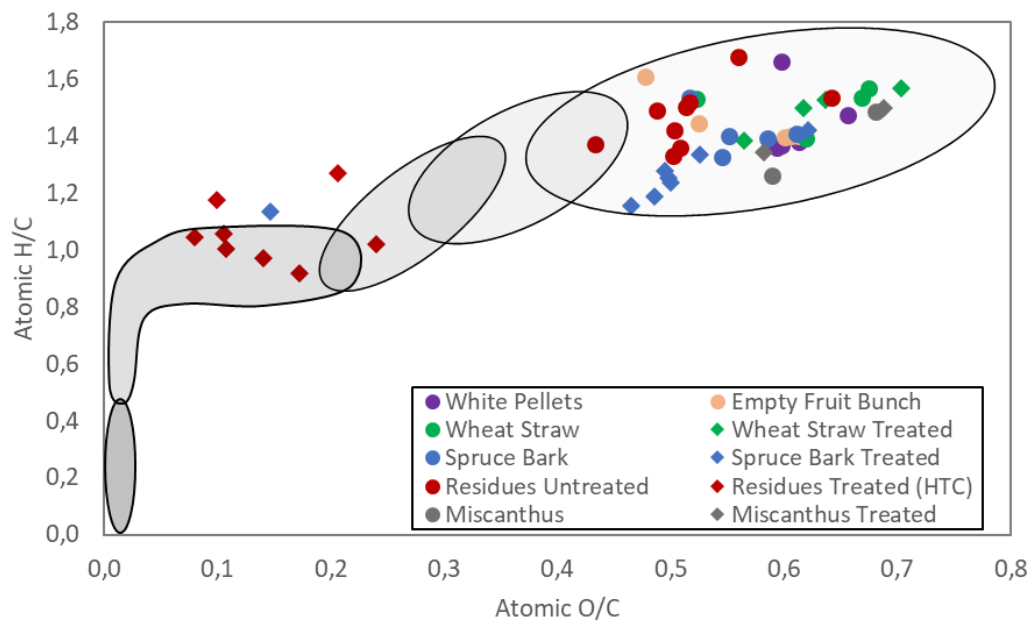


Figure 6: Van-Krevelen diagram of each of the datasets in the Bioefficiency database. The grey areas indicate the usual classification of fuels: (from bottom left to top right) anthracite, coal, lignite, peat, and biomass.

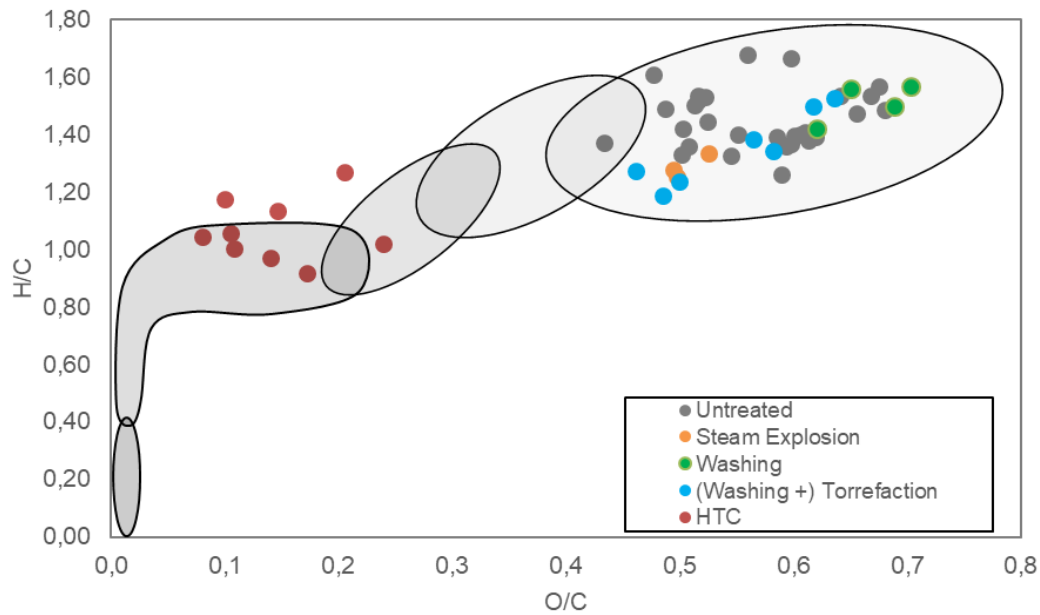


Figure 7: Van-Krevelen diagram re-ordered to show the influence of each pre-treatment method.

One of the other benefits of pre-treatment lies of course in the improved combustion behaviour and reduction of corrosive behaviour. To investigate this, a multitude of fuel indices are suggested in literature, mostly comparing XRF or ICP data of the fuel's ashes. Since most of these indices were designed with coal in mind as the main fuel, the transferability of these indices is often questionable. However, some classifications have been carried out focusing on biomasses [5–12]. Since this Deliverable does not focus on the evaluation of these indices, a selection of methods is instead presented based on literature. Vassilev et al. [5–7] proposed a classification based on the different mineral components of the biomass ashes, which can be seen in Figure 8. The authors suggest a classification into the following three major associations: Si-Al-Fe-Na-Ti, Ca-Mg-Mn and K-P-S-Cl. After this, each ash is separated into four different types (S, C, K and CK type) based on their oxide composition. For this, the mentioned metal oxides are normalised to a 100% basis and added to the diagram. This classification is useful for predicting the behaviour of the different elements during combustion. The interpretations on the fuel deposition behaviour vary, however. Vassilev et al. [6] suggest for example that fuels composed of mainly C-, CK- and K-types are highly volatile and will lead to low-melting ashes in the boiler, while the more acidic S-types are rather stable and high-melting. García et al. [12] suggest on the other hand that S- and K-types are most difficult due to silicate formation and presence of potassium. It can be observed that the wood pellets with added coal fly ash (WPA) are classified in the "coal area" of the diagram. It should be noted that the straw sample analysed here is not representative for general straw fuels, as it contains an unusually high amount of Si. Usual straw samples instead have considerably lower shares of Si and higher shares of K, therefore posing very high risks during combustion.

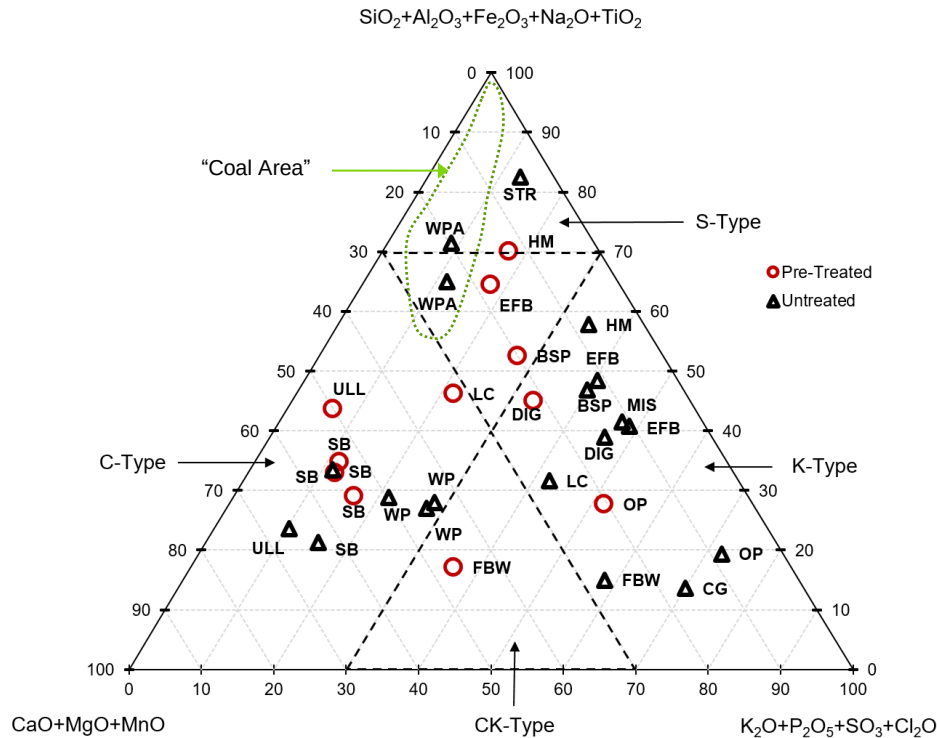


Figure 8: Ternary diagram for the classification of biomass fuels based on their mineral compositions [5–7]. The abbreviations for each fuel can be found at the beginning of this deliverable.

Another way of classifying biomasses is the separation into CaO (+MgO), SiO₂ and K₂O (+Na₂O) [9,10]. Näzeli et al. [9] suggested a categorisation based on the share of slag which is to be expected in each ash during combustion. Applying this categorisation to the Bioefficiency database results in Figure 9. When looking at this classification, it should be noted that the benefits of added Al-Si-species on the formation of aerosols is not captured correctly. For this reason, the "wood pellets + fly ash" sample is not included in this diagram.

However, most untreated residual biomasses are "correctly" labelled as difficult in this visualisation. This demonstrates that these classifications, while giving a useful first hint at a fuel's properties, can never truly capture all facets of a true in-depth analysis of each fuel. The reader is again reminded that the straw sample is not representative due to its high shares of Si.

A practical ash classification was carried out as part of the Bioefficiency project as well, and the results can be found in the respective ash-related deliverables.

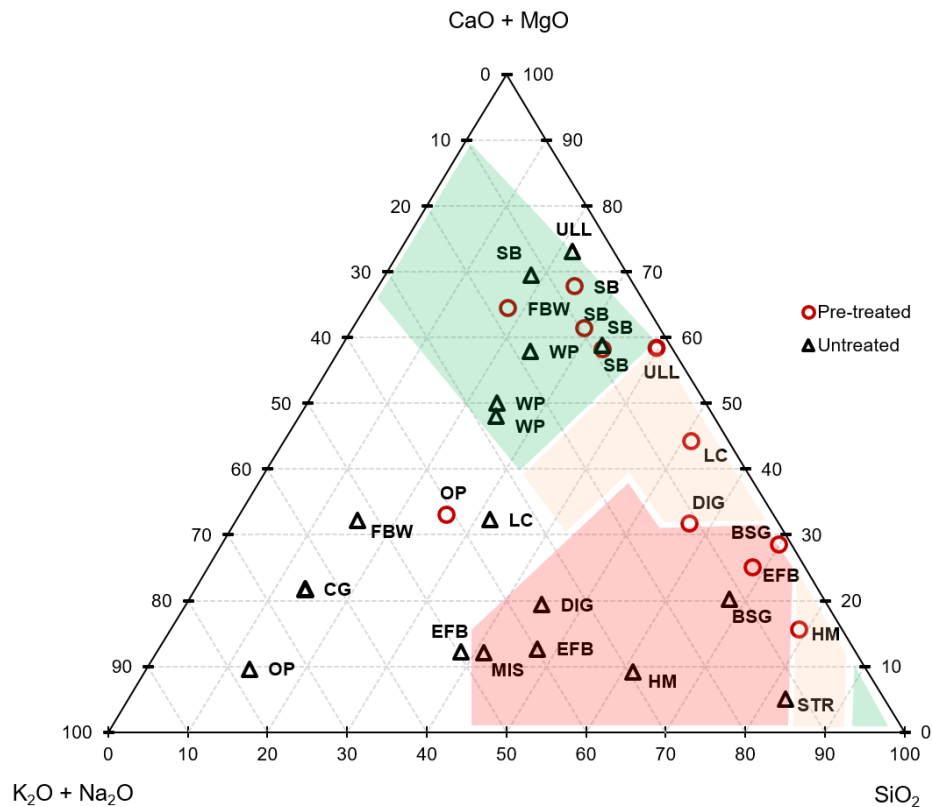


Figure 9: Ternary diagram for the classification of biomass fuels according to [10] with categorisation based on the fraction of fuel ash that forms slag [9]: green: < 7 wt.-%, yellow: 7 - 50 wt.-%, red: > 50 wt.-%. The abbreviations for each fuel can be found at the beginning of this deliverable.

Finally, one advantage of having a database of fuels analysed by multiple partners is the ability to compare different measurements of the same fuel by multiple partners. To investigate this, the standard deviations of these measurements were analysed in Table 1.

Table 1 shows that most proximate and ultimate analyses (moisture, ash content, C, H, N, O, S) show good repeatability, with standard deviations rarely exceeding 10% of the measured values. The XRF data was not normalised to 100% to compare the truly measured values of each machine. As the sum of all oxides sometimes is as low as 75 % and sometimes exceeds 110%, this adds an additional source of error that is inherent with the measurement method. However, the results still show relatively good repeatability, with standard deviations rarely exceeding 25 % of the measured values. Overall, the measurements never show strong discrepancies and are within the reasonable range of measurement errors given the method of comparison.

Table 1: Highlighting the repeatability of different in proximate, ultimate and XRF measurements. XRF measurements were not normalised to 100% beforehand. Not all measurements were always carried out by all partners; nmeasurements represents the total amount of different analyses.

Sample nmeasurements Partners	Wood Pellets 3 MPHS, TUM (2)	Wheat Straw 3 TUM, TNO, ABO	Spruce Bark 6 TUM (2), MHPS (2), VAL, TNO, ABO	Empty Fruit Bunch 5 TUM, MHPS, VAL (3)
H ₂ O (wt.-% ar)	5,90 ± 0,03	5,36 ± 2,77	7,72 ± 0,77	9,18 ± 1,60
Ash (wt.-% db)	1,32 ± 0,19	11,44 ± 1,03	3,95 ± 0,77	8,87 ± 0,18
C (wt.-% db)	50,38 ± 1,02	43,67 ± 1,20	51,14 ± 1,20	47,77 ± 2,06
H (wt.-% db)	6,32 ± 0,65	5,45 ± 0,19	6,02 ± 0,42	5,78 ± 0,60
O (wt.-% db)	41,77 ± 1,23	38,06 ± 0,75	38,25 ± 1,65	35,80 ± 2,41
N (wt.-% db)	0,13 ± 0,02	0,58 ± 0,15	0,41 ± 0,03	1,18 ± 0,13
S (wt.-% db)	0,07 ± 0,01	0,14 ± 0,05	0,08 ± 0,06	0,19 ± 0,08
Cl (wt.-% db)	0,01 ± 0,00	0,23 ± 0,08	0,01 ± 0,00	0,53 ± 0,03
Al ₂ O ₃ (wt.-% ash)	3,09 ± 0,62		1,97 ± 0,66	3,35 ± 0,77
CaO (wt.-% ash)	35,18 ± 5,86		44,17 ± 10,99	5,31 ± 1,83
Fe ₂ O ₃ (wt.-% ash)	2,13 ± 0,93		1,11 ± 0,16	3,10 ± 0,42
K ₂ O (wt.-% ash)	18,42 ± 3,40		7,15 ± 2,75	34,09 ± 5,92
MgO (wt.-% ash)	6,79 ± 0,18		3,94 ± 1,61	4,32 ± 1,73
MnO (wt.-% ash)	0,55 ± 0,48		1,89 ± 0,15	0,09 ± 0,02
Na ₂ O (wt.-% ash)	0,69 ± 0,03		0,67 ± 0,24	0,49 ± 0,15
P ₂ O ₅ (wt.-% ash)	3,31 ± 0,77		3,31 ± 1,71	3,70 ± 0,84
SiO ₂ (wt.-% ash)	19,51 ± 0,71		18,45 ± 4,74	33,11 ± 4,80
SO ₃ (wt.-% ash)	1,75 ± 0,59		1,00 ± 0,28	2,07 ± 0,47
TiO ₂ (wt.-% ash)	0,21 ± 0,08		0,22 ± 0,03	0,18 ± 0,03

4. Integration into the Public "Phyllis2" Database

One of Bioefficiency's project partners, ECN part of TNO, hosts a biofuel analysis database that is publicly available at <https://phyllis.nl/> (August 2019). To enhance visibility and accessibility, it was decided to upload all gathered analysis data to this database. The data was therefore forwarded to ECN part of TNO and was added to the database.

The data can be found in the Phyllis2 database by searching the relevant keywords (e.g. the name of the fuel). The Bioefficiency project name was added in the comments area of each fuel. Figure 11 shows the home page of the database.

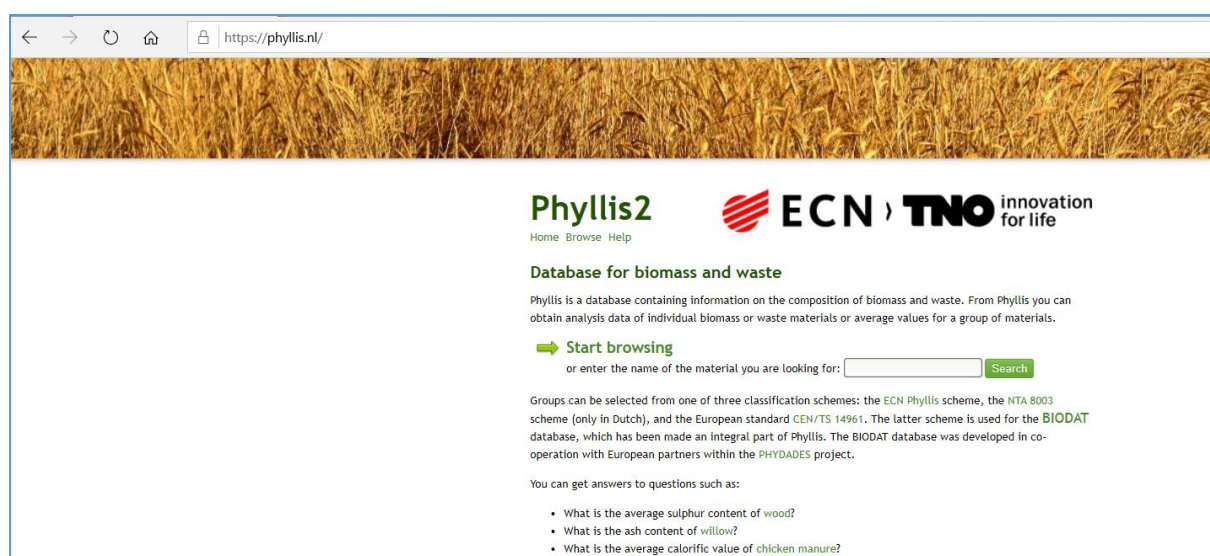


Figure 10: The homepage of the Phyllis2 Database

A search using e.g. the keyword “spruce bark” will deliver the total list of samples in the database. By selecting one of the samples on the left, the collected data for this sample is shown on the right. In Figure 12, the results of both options are shown.

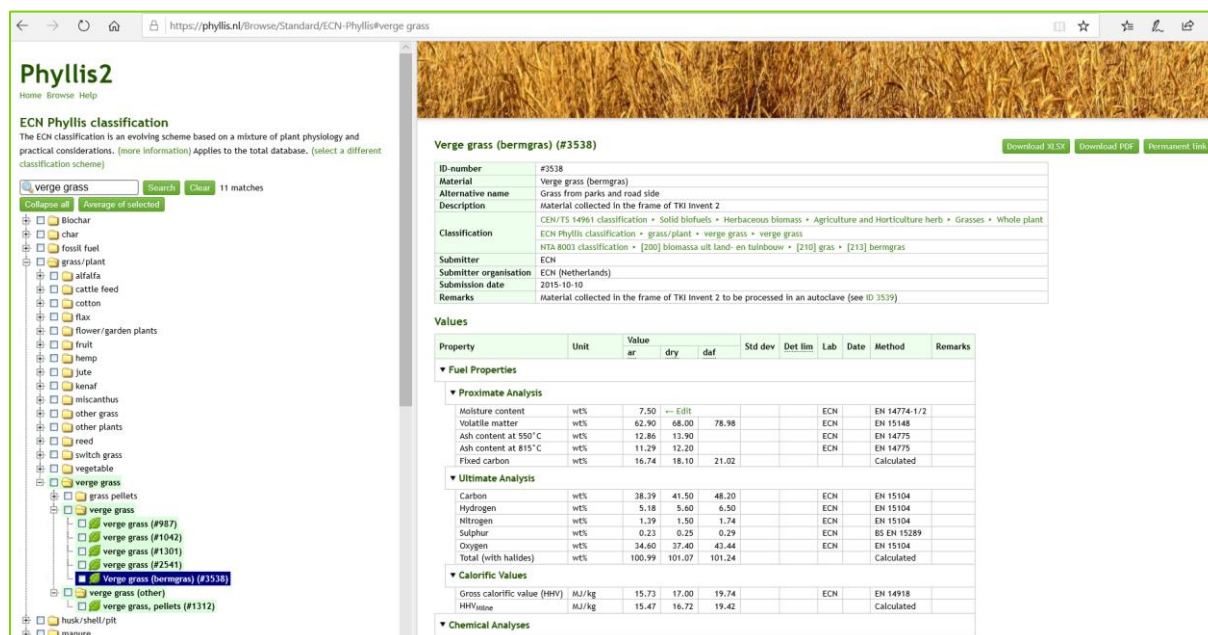


Figure 11 Results of the keyword search.

Results can also be averaged by selecting several samples and using the “average” button. The analysis data can be downloaded as an Excel file (.xlsx) or as a PDF file (average results are only available as Excel files).

Besides using the right keywords, the samples of the Bioefficiency project can be found by using the unique sample number as shown in the Table presented in the Annex (see column “Phyllis #”).

Supplying of new data will also be possible soon through the updated Phyllis2 site, as one of the objectives within the EU project BRISK2 (www.brisk2.eu). The new functionality of Phyllis2 will provide an upload file (.xlsx format) to fill in the data and send it by mail (Secretariaat-BEE@tno.nl) to ECN part of TNO. The data will then be checked and uploaded to the database.

5. Conclusion

The Bioefficiency project analysed 14 different fuels with different pre-treatments, generating a total of 57 different datasets, often containing multiple different analyses each. The data was entered into a centralised and simple database system, providing each partner with the opportunity to compare measurements and conclude new information about the different fuels and pre-treatments. This valuable data will serve and be available for the general public now and for the future through the upload to the open database Phyllis2.

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Annex

Name	Description	Phyllis #
White Pellets	Pretreatment: no	3558
Milled White Pellets	Pretreatment: no	3559
Milled White Pellets	Pretreatment: no	3560
Wheat Straw	Pretreatment: pulverised and sieved	3561
Wheat Straw	Pretreatment: no	3562
Wheat Straw	Pretreatment: no	3551
Wheat straw	Pretreatment: no	3563
Wheat Straw	Pretreatment: yes, washed	3552
Wheat Straw	Pretreatment: yes, washed + torrefied	3553
Wheat straw	Pretreatment: yes, torrefied	3564
Wheat Straw	Pretreatment: yes, washed and torrefied	3565
Spruce Bark	Pretreatment: no	3566
Spruce Bark Pellets	Pretreatment: no	3567
Spruce Bark	Pretreatment: no	3568
Spruce bark Pellets	Pretreatment: no	3569
Spruce Bark	Pretreatment: no	3554
Spruce Bark untreated	Pretreatment: no	3570
Spruce Bark	Pretreatment: yes, washed	3555
Spruce Bark	Pretreatment: yes, washed + torrefied	3556
Spruce Bark	Pretreatment: yes, torrefied	3571
Spruce Bark	Pretreatment: yes, washed and torrefied	3572
Spruce Bark	Pretreatment: yes, steam exploded	3573
Spruce Bark Pellets	Pretreatment: yes, Steam Explosion	3574
Spruce Bark Powder	Pretreatment: yes, Steam Explosion	3575
Spruce Bark Pellets	Pretreatment: yes, Steam Explosion	3576
Spruce Bark	Pretreatment: yes, HTC, 280°C, 4h	3577
Brewery Spent Grains	Pretreatment: no	3578
Brewery Spent Grains	Pretreatment: Yes, HTC, 280°C, 4h	3579
Empty Fruit Bunch	Pretreatment: no	3580
Empty Fruit Bunch	Pretreatment: no	3581
Empty Fruit Bunch	Pretreatment: no	3582
Empty Fruit Bunch	Pretreatment: no	3583
Empty Fruit Bunch	Pretreatment: no	3584
Empty Fruit Bunch	Pretreatment: yes, HTC, 4h, 280°C	3585
Olive Pomace	Pretreatment: no	3586
Olive Pomace	Pretreatment: yes, HTC, 270°C, 4h	3587
Fir Brushwood	Pretreatment: no	3588
Fir Brushwood	Pretreatment: yes, HTC, 240°C, 2h	3589
Urban Leaf Litter	Pretreatment: no	3590
Urban Leaf Litter	Pretreatment: yes, HTC, 270°C, 4h	3591
Coffee Grounds	Pretreatment: no	3592
Digestate	Pretreatment: no	3593
Digestate	Pretreatment: yes, HTC, 240°C, 4h	3594
Lawn Cuttings	Pretreatment: no	3595
Lawn Cuttings	Pretreatment: yes, HTC, 270°C, 4h	3596
Horse Manure	Pretreatment: no	3597
Horse Manure	Pretreatment: yes, HTC, 270°C, 4h	3598
Road-side grass	Pretreatment: no	3548
Road-side grass	Pretreatment: yes, washed	3549
Road-side grass	Pretreatment: yes, washed + torrefied	3550
Miscanthus	Pretreatment: no	3599
Miscanthus	Pretreatment: no	3547
Miscanthus	Pretreatment: yes, washed	3546
Miscanthus	Pretreatment: yes, washed + torrefied	3544